

Flushner: A 3D Printing Technique That Inserts Stepped Objects to Achieve Surface Uniformity and High Speed

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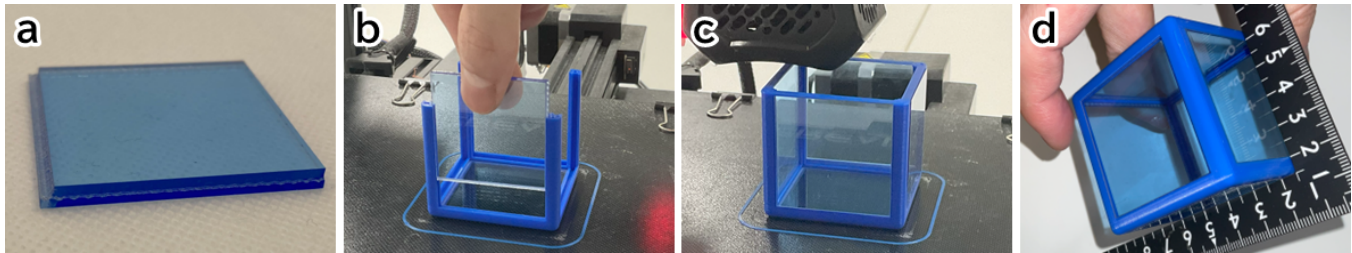


Figure 1: Flushner workflow. (a) prepare the stepped inserts, (b) pause the printing and then set the inserts in the 3D printed object, (c) restart the printing over the inserts, and (d) complete a cube with flush faces.

ABSTRACT

Rapid prototyping is a technique used to quickly materialize ideas in the early stages of product development. It has become commonplace with the widespread use of FDM 3D printers. However, a notable challenge exists that the print time is too long for rapid iteration. Several approaches have already been taken to address this issue. We propose a technique to further enhance 3D print speed by inserting stepped objects into the 3D object being printed by using the print-pause-print protocol. The inserted objects form surfaces flush with the 3D printed surfaces and the print time by the 3D printer for walls, infill, and support structures can be reduced. The technique does not require expensive or modified 3D printers and is very effective with cheap and simple single-extruder FDM 3D printers. Several application examples are implemented, and it is confirmed that the printing time is reduced by up to 89%.

CCS CONCEPTS

• Human-centered computing → Systems and tools for interaction design.

KEYWORDS

rapid prototyping, 3D printing, laser cutting

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1 INTRODUCTION

Rapid prototyping is a technique that can quickly realize ideas in the early stage of product development. The process involves transforming ideas into physical prototypes and iteratively refining and modifying the product's configuration. These prototypes are utilized to evaluate designs, collect feedback, and inspire new designs. Fused Deposition Modeling 3D printers (FDM 3D printers) are one of the most effective and commonly used tools for creating prototypes. Due to the recent dramatic decrease in cost, they are used both in industry and at home. They stimulate people's creativity and help people solve their own problems. However, a remaining challenge is that print times are still long relative to the size of the object. For example, it takes more than 8 hours to fabricate a cube as small as 5 cm on each side. This upsets the iteration process, which is the key to rapid prototyping. A lot of research has been conducted to try to resolve this issue.

In this paper, we propose Flushner, a technique that can be applied to inexpensive and simple home-use FDM 3D printers to reduce the printing time. It replaces surface elements of the 3D printed object with pre-built inserts to basically eliminate the time needed to print walls, infill, and support structures.

2 RELATED WORK

Several techniques have been proposed to rapidly create physical 3D objects. One of the primary techniques is to use laser cutters, which can cut and engrave flat plates quickly. For example, 3D objects are manually assembled from the cutting the cut plates [11][20]. Techniques to ease the difficulty of the assembly process have been proposed [1][2]. It is even possible to use the laser cutters to help assemble the components [16][25][18]. Pin-based shape displays can also be used for the rapid creation of 3D objects [24][29].

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While these techniques achieve rapid 3D object creation, all have some limitations; for example, curved and detailed surfaces are not supported and materials that can be used are limited.

Considering such limitations and the cost of implementation, FDM 3D printers are the first choice for creating 3D objects, especially for fabrication at home. FDM 3D printers extrude filament sequentially and, layer by layer, to create 3D objects. While objects consist of walls (surfaces) and infill, 3D printers additionally need to print support structures. Increasing the speed of the extruding filament is an effective way to reduce the total printing time, and some 3D printer manufacturers are competing to do so. Developing techniques to improve some of the printing processes is another approach to reducing the printing time. They are roughly divided into two categories; (1) modification of key factors including the model, slicer, and printer, and (2) incorporation of existing objects.

2.1 Modification of the model, slicer, and printer

Optimizing the parameters constituting the printed 3D object is beneficial in reducing the printing time. For example, WirePrint [15] and On-The-Fly Print [19] convert the 3D model into a mesh-like object. The printed object works as a low-fidelity wireframe preview [15]. Because they omit the printing of walls and infill and dramatically reduce printing time, they allow designers to iterate the process quickly. However, the printed objects are fragile and cannot be tested for physical interaction.

When printing a CAD-based 3D model as a physical object, a slicing process is required to create the tool path. The printing time depends on the slicing strategy. Techniques such as optimization of infill structures [28], reduction of wasted motion [14], and improving the support structures [8] reduce the printing time and the material used without degrading quality.

Dividing the 3D model into several different parts can reduce the printing time [12][30]. The idea is to reduce the infill and support structure. However, manual assembly is required after printing. As in the effect seen when the model is divided, increasing the number of print axes also decreases the infill and support structures [9].

2.2 Incorporating inserts

Numerous studies have focused on reducing printing time by incorporating inserts that replace regions that need to be printed. For example, TrussFab prints hubs to connect plastic bottles to fabricate large-scale structures [13]. Scrappy uses scrap material as infill [26]. faBrickation uses standard building blocks to substitute for subvolumes and prints only those parts that need to be high resolution [17]. Other research on the use of building blocks is found in [4][22]. Substiports uses reusable objects as ad hoc structural supports [27]. Instead of using existing objects available on the market, some studies have created dedicated objects for use in specific models. For example, CofiFab replaced the internal 3D printed infill with a structure that could be produced with a laser cutter [23]. Platener is one of the low-fidelity fabrication techniques but preserves the technical function of the object [3]. It reduces the fabrication time by dividing the roles between the 3D printer and the laser cutter, so the parts are built separately. These techniques are expected to be more effective in reducing print time than using

generic and reusable blocks. Unfortunately, a manual assembling process is required to complete the final object.

Incorporating existing objects not only reduces the printing time, but it can also be used to add some novel functionalities to the printed objects. For example, Medley embeds various materials to change the mechanical properties [5]. HotFlex [10] and Encore [6] can create functional objects. PrintGami [7] embeds electronics, while Liquido [21] creates water-filled objects.

There are two ways to print 3D objects around the use of existing objects; printing components and then assembling them manually and adopting the print-pause-print (PPP) protocol. For example, CofiFab [23] and Platener [3] take the former approach, while PrintGami [7] and Liquido [21] take the latter. The difference between the two is the timing of user intervention. With the PPP protocol, the 3D printer pauses and then the user manually inserts one or more objects before resuming the printer. This simplifies the assembly process and eliminates the additional parts, such as joints and inserts, that are needed in the former strategy to support manual assembly.

3 PROPOSED METHOD

We propose a technique to reduce printing time that supports even cheap single extruder 3D printers without any modification. It creates inserts that fit the target 3D object being printed. While inserted parts are one-offs as in [23] and [3], the technique uses the PPP protocol. The most unique characteristic of our proposal is that it can utilize flat surfaces without the need for model approximation, which means the inserts are flush with the 3D printed surfaces. When the inserts are made by a laser cutter, the time taken to print walls is reduced. Moreover, the time to print infill and supports can be reduced in some cases. It is possible to not only reduce the printing time but also enhance the aesthetic appearance of 3D-printed objects.

The workflow of the technique for creating a cube with rounded corners is shown in Figure 1. First, stepped inserts are created in advance, as shown in Figure 1 (a). When the 3D printer reaches the specified position, the 3D printer pauses, the user sets the insert (Figure 1 (b)), and printing is resumed (Figure 1 (c)). Finally, a cube is complete (Figure 1 (d)). In the case of the cube shown in Figure 1, three PPP events are coded: insert the bottom surface, the four sides, and the top. Although all six inserts are flat surfaces with a slightly concave edge on one side, as shown in Figure 1 (a), the way they are set in the 3D printing object is slightly different as follows.

- **Bottom surface** (Figure 2 (a)). Each insert has steps on all four edges. First, the 3D printer builds the outer frame holding the insert, red region in Figure 2 (a); its height equals the thickness of the insert. A lip is formed to prevent the insert from falling out, and its edge has a diagonal bevel to smoothly fit the insert; see the blank (white) between the red and blue. When the 3D printed reaches the desired height, the insert is set, blue in Figure 2 (a). The 3D print is then resumed. The inserted object and the 3D printed surface are smoothly connected to achieve surface uniformity.
- **Side surface** (Figure 2 (c), (d), and (e)). The inserts have steps in all edges except the upper edge. The 3D printed frame (four vertical posts) has grooves like a sliding door, which

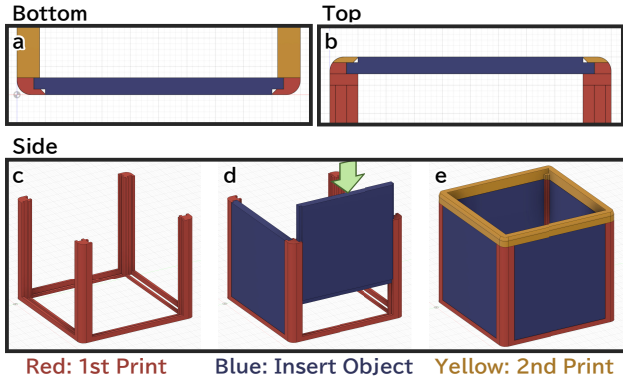


Figure 2: Setting procedure for (a) bottom, (b) top, and (c)-(e) side surfaces. Red represents the 3D printed region before setting, blue the insert, and yellow the 3D printed region after setting.

are designed to pinch the inserts, red in Figure 2 (c). The 3D printer prints up to the height of the insert and pauses when it reaches that height. The four inserts are then set, blue in Figure 2 (d). With the insert set, the 3D printing is resumed, yellow in Figure 2 (e).

- **Top surface** (Figure 2 (b)). The insert has steps in all four edges. The 3D printer prints up to the height of the lower step of the stepped object to be inserted and pauses, red in Figure 2 (b). The user manually sets the insert, blue in Figure 2 (b). After that, 3D printing is resumed, yellow in Figure 2 (b). Since the top of the insert is higher than the 3D printed surface at this moment, the nozzle path is designed to avoid hitting the insert. This is shown by the triangular spaces between the blue and yellow in Figure 2 (b).

4 IMPLEMENTATION

The 3D printer used was a Creality Ender3-v2 and PLA filament was used. The inserted objects were created by a laser cutter, FusionMaker 12-40W, using a 3 mm thick acrylic plates. The steps of the objects were made by the laser cutter's engraving function. The model was manually divided into those made by the 3D printer and those by the laser cutter, and the G-code was also manually altered. To confirm the proposed technique, we evaluated the reduction in time it took to print simple cubes. Several more complicated models were developed and tested using this implementation setup.

4.1 Evaluation

The time difference between Flushner and normal 3D printing was compared for the fabrication of simple cubes of different sizes, from 15 to 100 mm. In these examples, Flushner eliminated the need for infill as the top surface did not need supports.

The results are shown in the graph in Figure 3. Note that the time taken by Flushner includes the time for laser cutting and the manual setting process. When cube size is less than 20 mm, normal

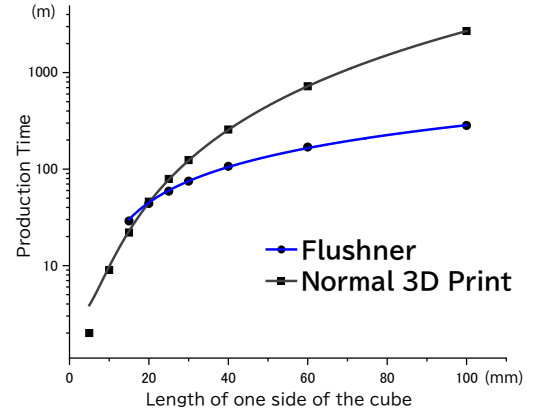


Figure 3: Creation time for Flushner and normal 3D printing. While Flushner print time was linear to the size of the cube, normal print time was proportional to cube size to the power of 2.6.

3D printing takes less time; at 25 mm or more, Flushner takes less time. For example, the creation time for cubes with 100 mm is reduced by 89%. In normal 3D printing, the printing time of the walls is approximately proportional to the square of the length, and the cube of infill area is proportional to the third power. The total time is their sum. On the other hand, the Flushner time is expected to be approximately proportional to the length. To confirm this, the measured plots were fitted with the following equation.

$$T = a + b * L^c \quad (1)$$

where T is the time to complete, a , b , c are variables, and L is cube size. Estimated c for normal 3D printing was 2.6, and 1.0 for Flushner in this case. These equations are shown in Figure 3 as curves.

4.2 Examples

Examples other than Figure 1 and their creation times are shown in Figure 4, (a) stacked boards using pillars, (b) slanted polyhedron, (c) basket, (d) hexagonal wine glass, (e) tablet stand, and (f) bridge. In examples (a) and (d), the horizontal inserts eliminate the need to print and then remove the support structures. They also show that both sides of the inserts can be used as surfaces. Examples (b) and (e) show Flushner also supports the case of slanted inserts. Examples (c) and (f) show that Flushner can eliminate the support structures for not only horizontal but also vertical inserts. While the time reduction rates for these examples are different from 18% to 79%, all show the effectiveness of the proposed technique. As shown, Flushner can greatly enhance the aesthetic appearance and yields completely flat surfaces unlike regular 3D printed surfaces.

5 DISCUSSION

Although the examples confirmed that the proposed technique reduces the creation time, there are some limitations. The most important one is that the current implementation lacks automatic

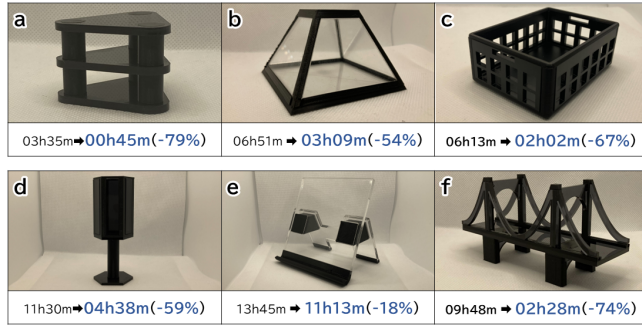


Figure 4: Examples developed, (a) stacked boards using pillars, (b) slanted polyhedron, (c) basket, (d) hexagonal wine glass, (e) tablet stand, and (f) bridge. The reduction in time is also shown, normal 3D print time on the left and Flushner time on the right.

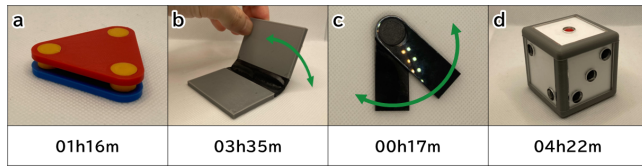


Figure 5: Implementations to explore the design space. (a) multicolor printing, (b) multimaterial printing (gray PLA and black TPU), (c) connecting solid objects (two acrylic plates are joined but they are rotatable), and (d) multicolor and multimaterial dice.

model transformation. The automatic process should include finding the regions that could be replaced, changing the original model, and modeling the inserts. The second one is that small gaps between the 3D printed surfaces and inserts can sometimes occur. This might be due to the machining accuracy, and some technique to eliminate these gaps may be required if quality is required.

5.1 Possible applications

Several implementations were tried to explore the design space of the proposed technique. The creations are shown in Figure 5, (a) multicolor printing, (b) multimaterial printing, (c) connecting solid objects, and (d) multimaterial and multicolor dice. Since it is possible to 3D print the inserts instead of laser cutting them, 3D objects with multicolor and/or multimaterial can be fabricated, such as (a), (b), and (d). The technique can be used to connect two rotatable parts, as in (c). Note that all of them achieved surface uniformity and they are created by the current setup with a simple single extruder 3D printer. Although printing time is not expected to be reduced, the technique makes it possible to create objects that have been difficult to create with cheap 3D printers at home.

6 CONCLUSION

A technique to speed up rapid prototyping by incorporating stepped inserts into the 3D printing process was proposed. The joints between the inserts and the 3D printed surfaces can be flush. An

experiment was conducted to evaluate the printing time of cubes. The results showed that the creation time with the technique is linear to cube size, it reduced the time needed by up to 89%. Moreover, several different implementation examples were successfully created to explore the proposal's design space.

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